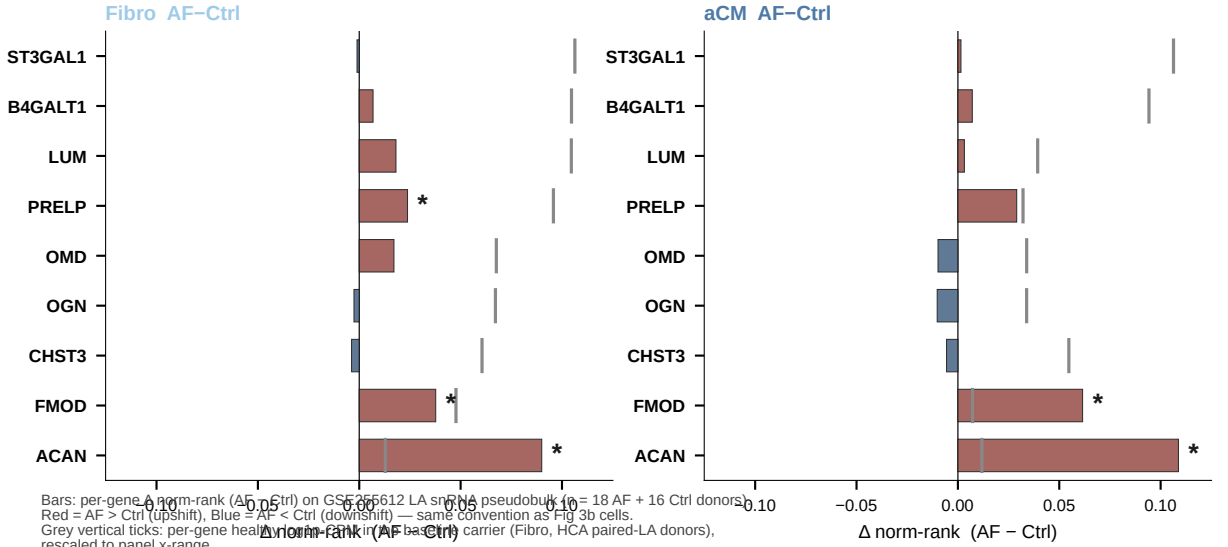
Fig 3e F16 Keratan sulfate biosynthesis — Cellular extension

Cellular extension



Bars: per-gene Δ norm-rank (AF − Ctrl) on GSE255612 LA snRNA pseudobulk (n = 18 AF + 16 Ctrl donors).
Red = AF > Ctrl (upshift), Blue = AF < Ctrl (downshift) — same convention as Fig 3b cells.
Grey vertical ticks: per-gene healthy-donor rank in the baseline carrier (Fibro, HCA paired-LA donors), rescaled to panel x-range.

Both compartments show concordant AF-time upshift (positive RED bars), in contrast to the predominantly downshift (blue) seen in the relay case (Fig 3d). The baseline carrier (fibroblast) is co-active with the neighbouring cardiomyocyte compartment; the bulk-detected change therefore reflects cellular extension — the carrier's activity change is amplified by adjacent cell types rather than re-routed.
Note that OGN, OMD and CHST3 show small divergent (negative) drift in atrialCM despite the overall upshift, indicating that extension is not a perfectly uniform pathway-wide event; the dominant extension signal is carried by ACAN, FMOD and PRELP (gene-level p < 0.05 in both compartments).

Significance markers: * = gene-level p < 0.05; . = gene-level p < 0.10 (donor permutation, n_perm = 5000).
Color note: cellular-fidelity badge colors (purple = relay, teal = extension) are decoupled from the carrier palette in Fig 3b to avoid cross-figure confusion.